SIMATS ENGINEERING
Department of Computer Science & Engineering
ASSESSMENT TOOL 2-COMPARATIVE ANALYSIS

Course Code:	**CSA12**	**Course Title:**	**Computer Architecture for Multicore Systems**
CO Assessed:	**CO4**	**Assessment:**	**ASSESSMENT TOOL2-COMPARATIVE ANALYSIS**
Weightage:	**30%**	**Total Marks:**	**25**
CO4 Statement:	*Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.*		
Student Name:		**Reg.No.:**	
Department:		**Date:**	

ANNEXURE A
COMPARATIVE ANALYSIS

1.Compare L1, L2, and L3 cache levels in terms of latency, bandwidth, and throughput, and analyze their impact on processor performance.
2.Differentiate SRAM and DRAM by evaluating their latency, bandwidth, and throughput characteristics in cache and main memory design.
3.Compare virtual memory with paging and cache memory mechanisms in terms of access latency and system throughput.
4.Analyze the differences between NAND Flash and DRAM with respect to latency, bandwidth, and suitability for hierarchical memory systems.
5.Compare MRAM and RRAM technologies by examining their performance in terms of access time, throughput, and energy efficiency.

Code of Conduct:

I V. Reddy Sreya *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

MARKS ALLOCATION

	Scope of Items (20%)	Comparison Depth (25%)	Use of Evidence (20%)	Critical Thinking (20%)	Clarity of Presentation (15%)
Q1					
Q2					
Q3					
Q4					
Q5					

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

L1, L2, and L3 Cache Comparison

Feature	L1 Cache	L2 Cache	L3 Cache
Location	Inside CPU core	Near/inside CPU core	Shared among cores
Size	Smallest	Medium	Largest
Latency	Very low	Low	Higher than L1/L2
Bandwidth	Highest	High	Lower than L1/L2
Throughput	Highest	High	Moderate
Access Speed	Fastest	Fast	Slower
Main Purpose	Frequently used data/instructions	Backup for L1	Shared backup for L1/L2

Comparison of L1, L2, and L3 Cache Levels

Parameter	L1 Cache	L2 Cache	L3 Cache
Latency	4–5 clock cycles (~1 ns)	10–14 clock cycles (~3–4 ns)	30–50 clock cycles (~10–15 ns)
Bandwidth	Highest (Terabytes/s per core)	High (~1–2 TB/s per core)	Moderate (~500–800 GB/s shared)
Throughput	Maximum (serves core instructions per cycle)	High (serves L1 misses)	Moderate (serves cross-core L2 misses)
Structure	Core-private, split (I-Cache / D-Cache)	Core-private, unified	Shared across cores, unified

L1, L2, and L3 Cache Levels Comparison

Parameter	L1 Cache	L2 Cache	L3 Cache
Latency	~1–4 cycles (fastest)	~10–20 cycles	~30–70 cycles
Size	16–64 KB (per core)	256 KB–1 MB (per core)	4–64 MB (shared)
Bandwidth	Highest (closest to core)	Moderate-high	Lower than L1/L2
Location	On-core, split I/D cache	On-core or per-core	Shared across cores
Throughput	Highest ops/cycle	Moderate	Lowest of the three

2. SRAM vs DRAM

Feature	SRAM	DRAM
Full Form	Static RAM	Dynamic RAM
Latency	Very low	Higher
Bandwidth	High	High, but generally lower effective latency
Throughput	High	High
Refresh	Not required	Required
Cost	Expensive	Less expensive
Density	Lower	Higher
Typical Use	Cache memory	Main memory

SRAM vs. DRAM in Cache and Main Memory Design

Characteristic	SRAM (Static RAM)	DRAM (Dynamic RAM)
Latency	Very Low (~1–10 ns)	High (~50–100 ns)
Bandwidth	Extremely High (bus directly integrated with CPU logic)	Moderate to High (limited by bus width and interface constraints)
Throughput	High per bit	High aggregate (burst modes like DDR5)
Cell Structure	6 transistors (6T) per bit; static bistable latch	1 transistor + 1 capacitor (1T1C); requires refresh cycles
Role in Hierarchy	Primary choice for On-Die Caches (L1, L2, L3)	Primary choice for Main System Memory

SRAM vs DRAM

Parameter	SRAM	DRAM
Cell structure	6-transistor flip-flop	1 transistor + 1 capacitor
Latency	Very low (ns range, no refresh needed)	Higher (needs row activation, refresh cycles)
Bandwidth	High, but limited by area/cost at scale	High aggregate bandwidth (wide buses, burst modes) but higher per-access latency
Throughput	High for random access patterns	High for sequential/burst access; lower for random access due to row-buffer misses
Density/Cost	Low density, expensive per bit	High density, cheap per bit
Power	Static power (leakage) dominant	Needs periodic refresh, adding dynamic power overhead

3. Virtual Memory with Paging vs Cache Memory

Feature	Virtual Memory/Paging	Cache Memory
Purpose	Extends available memory using storage	Speeds up CPU data access
Access Latency	Very high when page fault occurs	Very low
Storage	RAM + secondary storage	SRAM
Throughput	Lower during page faults	High
Speed	Slower	Much faster
Management	OS and memory management unit	Hardware/cache controller
Main Benefit	Allows large programs to run	Reduces CPU memory access time

Virtual Memory & Paging vs. Cache Memory

Feature	**Cache Memory System**	**Virtual Memory & Paging**
Access Latency	**Nanoseconds** (1 ns to 15 ns)	**Microseconds to Milliseconds** (Page Fault to SSD/HDD)
Throughput Impact	Prevents core starvation by keeping pipeline fed with fast SRAM hits.	Maximizes multi-programming throughput by allowing virtual address mapping beyond physical RAM bounds.
Mechanism	Hardware-managed (SRAM tagging and hit/miss logic).	Hardware + OS-managed (TLB, Page Tables, Page Fault Handling).
Unit of Transfer	Cache Line (typically 64 Bytes).	Page Size (typically 4 KB to 2 MB/1 GB huge pages).

Virtual Memory (Paging) vs Cache Memory

Aspect	Virtual Memory / Paging	Cache Memory
Purpose	Extends addressable memory using disk/SSD as backing store; provides isolation	Reduces average latency to frequently used data from a slower main memory
Access latency on hit	TLB hit: near-zero overhead; page table walk on TLB miss: tens of cycles	Cache hit: single-digit to tens of cycles
Access latency on miss	Page fault: milliseconds (disk/SSD I/O)	Cache miss: goes to next level, tens to hundreds of cycles
Management	OS-managed (page tables, replacement policies like LRU/Clock)	Hardware-managed (replacement policies in hardware, transparent to software)
Granularity	Pages (typically 4 KB, or larger with huge pages)	Cache lines (typically 64 bytes)
Throughput impact	A page fault can stall a process for millions of cycles; amortized over long-running programs it's usually rare	Cache misses are frequent but individually cheap; aggregate effect dominates average CPI

4. NAND Flash vs DRAM

Feature	NAND Flash	DRAM
Type	Non-volatile	Volatile
Latency	Higher	Lower
Bandwidth	High for sequential operations	Very high
Throughput	High, especially for storage	Very high for memory operations
Data Retention	Retains data without power	Loses data when power is removed
Refresh	Not required	Required
Main Use	SSDs, USB drives, storage	Main memory
Hierarchy Level	Secondary storage	Main memory

NAND Flash vs. DRAM

Attribute	DRAM (Main Memory)	NAND Flash (Secondary Storage)
Latency	~50–100 ns	~10–100 μs (Reads), ~1–10 ms (Writes)
Bandwidth	Very High (e.g., DDR5 up to 64 GB/s per channel)	Moderate (PCIe Gen 5 NVMe up to 14 GB/s)
Volatility	Volatile (loses data on power down)	Non-Volatile (retains data)
Access Granularity	Byte/Word addressable	Block/Page addressable (Requires erase-before-write)

NAND Flash vs DRAM

Parameter	**NAND Flash**	**DRAM**
Latency	Microseconds (read ~25–100 μs including controller overhead; write/erase far slower)	Nanoseconds (tens of ns)
Bandwidth	Lower per-die, but aggregated across channels/dies in SSDs can be high for sequential I/O	Very high (GB/s per DIMM, tens of GB/s per channel)
Endurance	Limited program/erase cycles (wear-out); requires wear leveling	Effectively unlimited write endurance
Volatility	Non-volatile (retains data without power)	Volatile (requires refresh, loses data on power-off)
Density/Cost	Very high density, low cost per bit	Lower density than NAND, higher cost per bit
Access granularity	Page-based reads, block-based erases (asymmetric read/write)	Byte/word addressable

5. MRAM vs RRAM

Feature	MRAM	RRAM
Full Form	Magnetoresistive RAM	Resistive RAM
Access Time	Very low	Very low
Latency	Low	Low
Throughput	High	High
Volatility	Non-volatile	Non-volatile
Energy Efficiency	High	Very high potential
Endurance	Very high	High
Main Principle	Magnetic states	Resistance states
Applications	Embedded memory, cache	Embedded memory, storage, AI hardware

MRAM vs. RRAM (Emerging NVM Technologies)

Performance Parameter	**MRAM (Magnetoresistive RAM - STT/SOT)**	**RRAM (Resistive RAM / ReRAM)**
Access Time / Latency	**Fastest** (~1–10 ns for reads/writes)	**Moderate** (~10–50 ns reads, slower writes)
Throughput	High (comparable to SRAM speeds)	High for read-heavy vector-matrix operations
Energy Efficiency	**High** (Low dynamic write energy, zero static leakage)	**High** (Ideal for low-power non-volatile computing)
Working Principle	Magnetic Tunnel Junction (MTJ) spin-state resistance	Conductive filament formation in dielectric material
Primary Use Cases	Embedded SRAM/L3 cache replacement, persistent RAM	In-memory computing, Neuromorphic/AI accelerators

MRAM vs RRAM

Parameter	MRAM (Magnetoresistive RAM)	RRAM (Resistive RAM)
Storage mechanism	Magnetic tunnel junction (MTJ), stores state via electron spin orientation	Resistance change in a dielectric material via formation/rupture of conductive filaments
Access time	Fast (comparable to SRAM in STT-MRAM variants, ~ns range)	Fast, but generally slightly slower/more variable than MRAM; read/write asymmetry
Throughput	High, supports high endurance cycling well suited to frequent read/write	Good, though switching variability can limit sustained throughput
Endurance	Very high (10^12+ cycles for STT-MRAM), approaching DRAM/SRAM levels	Moderate-high, generally lower endurance than MRAM (10^6–10^10 cycles depending on material)
Energy efficiency	Low read energy; write energy moderate (current-driven switching)	Very low read/write energy in ideal cases, but filament formation can require higher initial "forming" voltage
Non-volatility	Yes	Yes
Scalability/Density	Historically lower density than RRAM due to MTJ cell size	Simpler cell structure (can be denser, crossbar-compatible), better scaling potential